

WASIM AKHTAR

B.Tech. Computer Science & Technology (AI & Machine Learning)

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EDUCATION

Presidency University, Bengaluru · *B.Tech. CSE (AI & ML)*

Sep 2023 – Jun/Jul 2027 (Expected)

CGPA: 6.73/10 (through Semester 6) · Relevant coursework: Artificial Intelligence & Machine Learning · Deep Learning & Reinforcement Learning · Explainable AI · Edge AI & IoT Analytics · Operating Systems · Data Structures & Algorithms · Database Management Systems · Cryptography & Network Security

RESEARCH EXPERIENCE

Independent Research — Adaptive Computing

akhwasim.netlify.app/research/adaptive-computing.html · Sep 2026 – Present · Self-directed

- Exploring how a software layer above the OS could observe user context and recurring workflows, and adapt the computing environment around them while remaining predictable, transparent, and under explicit user control.
- Builds directly on the pattern-learning work in Luna; currently in the question-forming stage, with open questions published and iterated publicly.

Student Researcher — Reinforcement Learning & Optimization · *Presidency University*

2025 – Present

- Contributed to the implementation and evaluation of Proximal Policy Optimization (PPO) for navigation in a $5,058 \times 5,058$ lunar-grid environment, and to the experimental analysis of learning performance; engaged with the methodological questions raised when initial results did not hold up under further scrutiny.

PROJECTS

Luna — Standalone Intelligent Terminal (Rust)

github.com/akhwasim/luna · MIT licence · v0.1.2 · 2★ on GitHub · Apr 2026 – Present · built in public on x.com/akhwasim

- Designed and shipped a standalone intelligent terminal in Rust: eight modules covering the shell REPL, command execution, AI provider dispatch, persistent memory, pattern learning, configuration, and a deterministic safety layer.
- Built a deterministic four-tier command-risk classifier (CRITICAL → LOW) in Rust that runs on every command path, including AI-suggested and workflow-step commands; safety is decoupled from the language model by design, with recursive shell-payload unwrapping to catch wrapped dangerous commands.
- Persistent SQLite-backed memory stores every command, error, workflow, and accepted/rejected AI suggestion; a learning loop re-ranks AI suggestions by the user's actual command history.

Sentinel — Software Supply-Chain Security Analyzer (Python)

github.com/akhwasim/sentinel · MIT licence · v1.0 · 7th-semester academic mini-project, Presidency University · Aug 2026 – Present

- Built a software supply-chain security analyzer in Python that generates CycloneDX 1.5 SBOMs, cross-references vulnerabilities against OSV.dev and the CISA Known Exploited Vulnerabilities (KEV) catalog, and flags packages with public exploits.
- Implemented AST-based reachability analysis to distinguish declared-but-unused dependencies from those actively imported in the codebase, reducing false-positive risk alerts.

- Designed a deterministic 0–100 supply-chain risk score with a fully documented rule set; scans without a lockfile are explicitly labelled DEGRADED rather than reported as complete.

Compass — Open-Source Issue Recommendation Platform

github.com/akhwasim/compass · MIT licence · v1.0 · live at compass-oss.netlify.app · Jul 2026 – Sep 2026 · React + FastAPI + Docker + GitHub Actions

- Built and deployed a full-stack platform (React + TypeScript on Netlify, FastAPI on Render) that recommends beginner-friendly GitHub issues by combining a deterministic confidence model with bounded LLM reasoning.
- Confidence model scores each issue on skill overlap, complexity, and repository health; the LLM is restricted to downgrading confidence on unclear issues and reasoning over verified issue text, never upgrading or inventing.
- Implemented a verification step that checks candidate file paths against the actual repository structure before display, after earlier iterations surfaced that LLMs would otherwise invent plausible-but-nonexistent paths.

LexRAG — Grounded Document Question-Answering API (Python)

github.com/akhwasim/LexRAG · Python · LangGraph · Pinecone · Gemini + Groq

- Built a document question-answering API with retrieval-augmented generation; returns citations with every grounded answer and refuses to answer when the corpus does not contain sufficient evidence.
- Implemented section-based chunking suited to short, structured documents, with deterministic chunk IDs so repeated ingestion overwrites existing vectors instead of creating duplicates.
- Routed the pipeline through a LangGraph state machine with a grade-and-branch flow (sufficient evidence → answer; otherwise → refusal); split embedding and generation across Gemini and Groq to stay within free-tier limits.

TECHNICAL SKILLS

Programming: Python · Rust · TypeScript · SQL

AI / Machine Learning: Machine Learning · Deep Learning · Reinforcement Learning · LLM APIs · LangGraph · RAG · Prompt Engineering · LLM System Design

Software Development: React · FastAPI · REST APIs · SQLite · PostgreSQL · Git · GitHub · Docker · Linux

Security / AI Systems: SBOM (CycloneDX) · Dependency Analysis · Static Analysis · OSV · CISA KEV · Supply-Chain Risk Scoring

LANGUAGES

Hindi (native) · English (full professional proficiency)